



Trepanation was a gruesome practice which involved making or scraping a hole in the human skull.



Bloodletting was another ancient practice which involved surgical incisions to make a person bleed, releasing bad blood to cure them.

Cutting you open

From painful, barbaric, often inexplicable practices, to the extreme precision of modern day, the evolution of surgery deserves an extended consultation

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Despite persistent advancements in medical science, the prospect of going under the knife still terrifies a large number of us. Not to aggravate those fears, but think about a time when none of the modern advancements was available to doctors and patients. A time when things like anaesthesia and bacteria-free environments weren't a standard in healthcare - does that scare you away even more? Then brace yourself, as we travel back to prehistoric times to understand how the reliability of surgery has reached its current status.

Ancient (12000 BC to 1 BC)

One of the earliest known surgical procedures was trepanation - literally drilling a hole in your head. According to evidence from one burial site in France, this procedure goes as far back as 6,500 BC according to some discoveries and 12,000 BC according to some. Carried out by a trephine, which refers to a class of instruments used for the cutting purpose, the procedure carried on even until the Middle Ages. The primary objective, according to the discovered records, seemed to be to relieve intracranial pressure. While there is no evidence if it fulfilled that purpose, there is some to support that a number of people subjected to it did survive.

While technically not surgery in most cases, examples of bone setting and splitting can be found in archaeological records, related to civilisations like the Aztecs. Another ancient surgical practice was that of bloodletting, practised among the Mesopotamians, the Egyptians, the Greeks, the Mayans, and the Aztecs. This practice would involve voluntarily bleeding out or extracting blood from the body, attributing it to various beliefs. One of those beliefs was that our bodily fluids, or "humours", had to remain in balance. Egyptians

even got inspired from the bleeding hippopotamus, not knowing that its sweat was red. While modern anaesthesia was still ages away, material extracted from herbs like opium and cannabis, and even alcohol was used to numb the senses of the patient and dull their perception. However, not being good enough, afflicted patients sometimes did choose to succumb to their ailments or continue suffering rather than undergo the surgery, and those who did also often succumbed to overdose from the numbing agents.

Interestingly, Ancient Egyptians gained extensive knowledge of the human anatomy from the practice of making mummies. Egyptian surgeons were also known to use clamps, sutures, and even cauterisation during their treatments. They also employed instruments like probes, saws, forceps, scalpels and scissors. They also used naturally occurring material like honey and willow bark as antiseptics.

Around the same time period as the Egyptians, Greeks also helped advance surgery significantly. They put wine (alcohol) into use as a disinfectant, and Herophilus is still sometimes referred to as the "Father of Anatomy". Our own culture wasn't far behind. Sushruta is known as the "founding father of surgery", and his period is estimated between 1200 BC - 600 BC. Much of what is known about him comes from the Sushruta Samhita, a set of Sanskrit volumes authored by him. These texts describe, in detail, the examination, diagnosis, treatment, and prognosis of numerous ailments, as well as procedures on performing various forms of cosmetic surgery, plastic surgery and rhinoplasty.

Historic (Up to the 18th Century)

Al-Zahrawi was one of the most influential physicians in the Middle Ages, notably around 900 AD. In his time, he authored the Kitab al-Tasrif, a thirty-volume encyclopedia of medical practices. In a Latin translation, the surgery chapter



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The advent of proper pain-relief led to the rise in number of surgeries and also the complexity involved in the process.